

Comparisons and asymptotic estimates



Outline

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Introduction: Comparisons and asymptotic estimates

- ▶ In this section we learn to **compare** two infinities or two infinitesimals by studying the limit of their ratio
- ▶ We introduce the **asymptotic** relation $a_n \sim b_n$, that is $\frac{a_n}{b_n} \rightarrow 1$
- ▶ **Asymptotic estimates** allow us to replace a complicated sequence with a simpler one with the same behaviour: a very powerful tool for computing limits

- ▶ We have seen that a sequence tending to 0 is an **infinitesimal**; a sequence that diverges (to $+\infty$, to $-\infty$) is called an **infinity** (infinite quantity).
- ▶ When two sequences are both infinitesimals or both infinities, it is useful to be able to compare them, to understand which of the two tends “**more rapidly**” to 0 or to infinity.

Example: Infinities

Examples of infinities are the following sequences:

$$\{\log n\}, \quad \{\sqrt{n}\}, \quad \{n^2\}, \quad \{2^n\}$$

Example: Infinitesimals

Examples of infinitesimals are the following sequences:

$$\left\{ \frac{1}{\log n} \right\}, \quad \left\{ \frac{1}{\sqrt{n}} \right\}, \quad \left\{ \frac{1}{n^2} \right\}, \quad \left\{ \frac{1}{2^n} \right\}$$

- Let $\{a_n\}$ and $\{b_n\}$ be **two infinities**; considering the limit of their ratio we have 4 cases:

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \begin{cases} 0 & \text{case 1)} \\ l \in \mathbb{R}, l \neq 0 & \text{case 2)} \\ \pm\infty & \text{case 3)} \\ \text{nonexistent} & \text{case 4)} \end{cases}$$

- 1) $\{a_n\}$ is an infinity of **lower order** than $\{b_n\}$
- 2) $\{a_n\}$ and $\{b_n\}$ are infinities of the **same order**
- 3) $\{a_n\}$ is an infinity of **higher order** than $\{b_n\}$
- 4) $\{a_n\}$ and $\{b_n\}$ are not comparable

- Let $\{a_n\}$ and $\{b_n\}$ be **two infinitesimals** (with b_n eventually different from zero); considering the limit of their ratio we have 4 cases:

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \begin{cases} 0 & \text{case 1)} \\ l \in \mathbb{R}, l \neq 0 & \text{case 2)} \\ \pm\infty & \text{case 3)} \\ \text{nonexistent} & \text{case 4)} \end{cases}$$

- 1) $\{a_n\}$ is an infinitesimal of **higher order** than $\{b_n\}$
- 2) $\{a_n\}$ and $\{b_n\}$ are infinitesimals of the **same order**
- 3) $\{a_n\}$ is an infinitesimal of **lower order** than $\{b_n\}$
- 4) $\{a_n\}$ and $\{b_n\}$ are not comparable

Comparisons and asymptotic estimates

- The case

$$\frac{a_n}{b_n} \rightarrow 1$$

is particularly important: in this case we say that the two sequences $\{a_n\}$ and $\{b_n\}$ are **asymptotic** (asymptotically equivalent).

- To indicate this, we write

$$a_n \sim b_n$$

(read: a_n is asymptotic to b_n)

- ▶ The asymptotic symbol is very useful in computing limits, thanks to the following **properties**:

Proposition 1: Asymptotic behavior

1. If $a_n \sim b_n$, the two sequences have the same behavior:
 - ▶ either they converge to the same limit,
 - ▶ or they both diverge to $\pm\infty$,
 - ▶ or neither of them has a limit.
2. We can write chains of asymptotic relations, i.e.:

$$\text{if } a_n \sim b_n \sim \dots \sim c_n \quad \text{then} \quad a_n \sim c_n$$

3. An expression made of a product or quotient of several factors can be estimated factor by factor:

$$\text{if } a_n \sim a'_n, b_n \sim b'_n, c_n \sim c'_n \quad \text{then} \quad \frac{a_n b_n}{c_n} \sim \frac{a'_n b'_n}{c'_n}$$

- ▶ **Warning:** the same does not hold for sums or for exponentials.

Proof (1/4)

1. We prove the first statement

if $a_n \sim b_n$ then $\{a_n\}$ and $\{b_n\}$ have the same behavior

► If $a_n \rightarrow l \in \mathbb{R}$, since

$$b_n = \frac{b_n}{a_n} \cdot a_n \quad \text{and} \quad \frac{b_n}{a_n} \rightarrow 1 \quad (\text{by definition of asymptotic}),$$

then by the theorem on the algebra of limits we have

$$b_n \rightarrow 1 \cdot l = l.$$

Proof (2/4)

- ▶ With the same steps, the theorem on the partial arithmetization of the infinity symbol allows us to conclude that

$$\text{if } a_n \rightarrow \pm\infty \text{ and } a_n \sim b_n \text{ then } b_n \rightarrow \pm\infty.$$

Observing that the asymptotic relation is symmetric, what we have just proved also shows that if $\{b_n\}$ converges (diverges), then $\{a_n\}$ converges (diverges) as well.

Proof (3/4)

- ▶ We conclude that if $\{a_n\}$ is irregular, then $\{b_n\}$ is irregular as well, because if, by contradiction, it were not, then by what we have just proved $\{a_n\}$ would also be convergent or divergent.

2. We prove the transitivity of the asymptotic relation:

$$\text{if } a_n \sim b_n \sim c_n \quad \text{then} \quad a_n \sim c_n.$$

Proof (4/4)

The hypotheses mean that

$$\frac{a_n}{b_n} \rightarrow 1 \quad \text{and} \quad \frac{b_n}{c_n} \rightarrow 1$$

Then by the theorem on the algebra of limits we have

$$\frac{a_n}{c_n} = \frac{a_n}{b_n} \cdot \frac{b_n}{c_n} \rightarrow 1.$$

3. The third property is proved analogously.



Example: Asymptotic sequences by applying the definition

We prove that:

$$\log n \sim \log(n+1)$$

Factoring out n in the second sequence we obtain

$$\log(n+1) = \log\left(n \left(1 + \frac{1}{n}\right)\right) = \log n + \log\left(1 + \frac{1}{n}\right)$$

Hence

$$\lim_{n \rightarrow +\infty} \frac{\log n}{\log(n+1)} = \lim_{n \rightarrow +\infty} \frac{\log n}{\log n + \underbrace{\log\left(1 + \frac{1}{n}\right)}_{\rightarrow 0}} = 1$$

Comparisons and asymptotic estimates

- ▶ A typical way to show that

$$a_n \sim b_n$$

consists in writing

$$a_n = b_n c_n \quad \text{with} \quad c_n \rightarrow 1.$$

That is, decomposing $\{a_n\}$ into the product of a sequence $\{b_n\}$ and a sequence $\{c_n\}$ that tends to 1.

Example: Asymptotic sequences via the decomposition method

Example:

$$\underbrace{2n^2 + 3n + 1}_{a_n} = \underbrace{2n^2}_{b_n} \underbrace{\left(1 + \frac{3}{2n} + \frac{1}{2n^2}\right)}_{c_n} \sim 2n^2 \quad \text{hence } a_n \sim b_n$$

since

$$\underbrace{\left(1 + \frac{3}{2n} + \frac{1}{2n^2}\right)}_{c_n} \rightarrow 1$$

Example: Computing limits with asymptotic estimates (1/3)

We compute the limit

$$\lim_{n \rightarrow +\infty} \frac{2n^3 + 4n + 1}{5(n+1)^3} = \left[\frac{\infty}{\infty} \right] \quad (\text{indeterminate form})$$

Proceeding as before we can write

$$\underbrace{2n^3 + 4n + 1}_{a_n} = \underbrace{2n^3}_{a'_n} \underbrace{\left(1 + \frac{2}{n^2} + \frac{1}{2n^3}\right)}_{\rightarrow 1} \sim 2n^3$$

$$5(n+1)^3 = \underbrace{5(n^3 + 3n^2 + 3n + 1)}_{c_n} = \underbrace{5n^3}_{c'_n} \underbrace{\left(1 + \frac{3}{n} + \frac{3}{n^2} + \frac{1}{n^3}\right)}_{\rightarrow 1} \sim 5n^3$$

Example: Computing limits with asymptotic estimates (2/3)

Using point 3 of Proposition on asymptotic behavior we can write:

$$\text{if } a_n \sim a'_n, c_n \sim c'_n \quad \text{then} \quad \frac{a_n}{c_n} \sim \frac{a'_n}{c'_n}$$

and obtain

$$\frac{2n^3 + 4n + 1}{5(n+1)^3} \sim \frac{2n^3}{5n^3}$$

that is, the two sequences have the same behavior. Hence

Example: Computing limits with asymptotic estimates (3/3)

$$\lim_{n \rightarrow +\infty} \frac{2n^3 + 4n + 1}{5(n+1)^3} = \lim_{n \rightarrow +\infty} \frac{2n^3}{5n^3} = \frac{2}{5}$$

Comparisons and asymptotic estimates

- ▶ Another way to show that

$$a_n \sim b_n$$

is to use the **substitution principle**.

- For example, knowing that

$$\lim_{n \rightarrow +\infty} \underbrace{\log n}_{a_n} \rightarrow +\infty$$

we can state

$$\lim_{n \rightarrow +\infty} \log c_n \rightarrow +\infty$$

where $\{c_n\}$ is any sequence diverging to $+\infty$, hence

$$\underbrace{\log n}_{a_n} \sim \underbrace{\log c_n}_{b_n}$$

Example: Computing limits with the substitution principle and asymptotic estimates (1/2)

We compute the limit

$$\lim_{n \rightarrow +\infty} \log_3(n^2 + 4n + 1)$$

using the substitution principle and defining

$$c_n = n^2 + 4n + 1 \quad \text{we have} \quad \lim_{n \rightarrow +\infty} c_n = +\infty$$

then

$$\log_3 n \sim \log_3(n^2 + 4n + 1)$$

Example: Computing limits with the substitution principle and asymptotic estimates (2/2)

and hence

$$\lim_{n \rightarrow +\infty} \log_3(n^2 + 4n + 1) = \lim_{n \rightarrow +\infty} \log_3 n = +\infty$$

Comparisons and asymptotic estimates

The fact that the asymptotic relation satisfies the 3 properties:

1. *Reflexive*: $a_n \sim a_n$
2. *Symmetric*: if $a_n \sim b_n$ then $b_n \sim a_n$
3. *Transitive*: if $a_n \sim b_n$ and $b_n \sim c_n$ then $a_n \sim c_n$

means that “**asymptotic**” is an **equivalence relation**.

Summary (1/2)

Comparisons:

- ▶ The limit of $\frac{a_n}{b_n}$ establishes whether two infinities (or infinitesimals) are of lower, equal or higher order, or not comparable

Asymptotic:

- ▶ $a_n \sim b_n$ if $\frac{a_n}{b_n} \rightarrow 1$; it is an equivalence relation
- ▶ Asymptotic sequences have the same behaviour (same limit or both irregular)
- ▶ Asymptotic estimates can be made factor by factor in products and quotients, **not** in sums or exponentials

Summary (2/2)

Methods:

- ▶ Definition: check that $\frac{a_n}{b_n} \rightarrow 1$
- ▶ Decomposition: write $a_n = b_n c_n$ with $c_n \rightarrow 1$
- ▶ Substitution principle: if $c_n \rightarrow +\infty$ and $\lim f(n) = L$, then $\lim f(c_n) = L$

Summary

- ▶ Two infinities or two infinitesimals are compared through the limit of their **ratio**
- ▶ $a_n \sim b_n$ (asymptotic) if $\frac{a_n}{b_n} \rightarrow 1$: they have the same behaviour
- ▶ In products and quotients each factor can be replaced by an asymptotic one, but not in sums
- ▶ Decomposition and the substitution principle make asymptotic estimates the main tool for computing limits